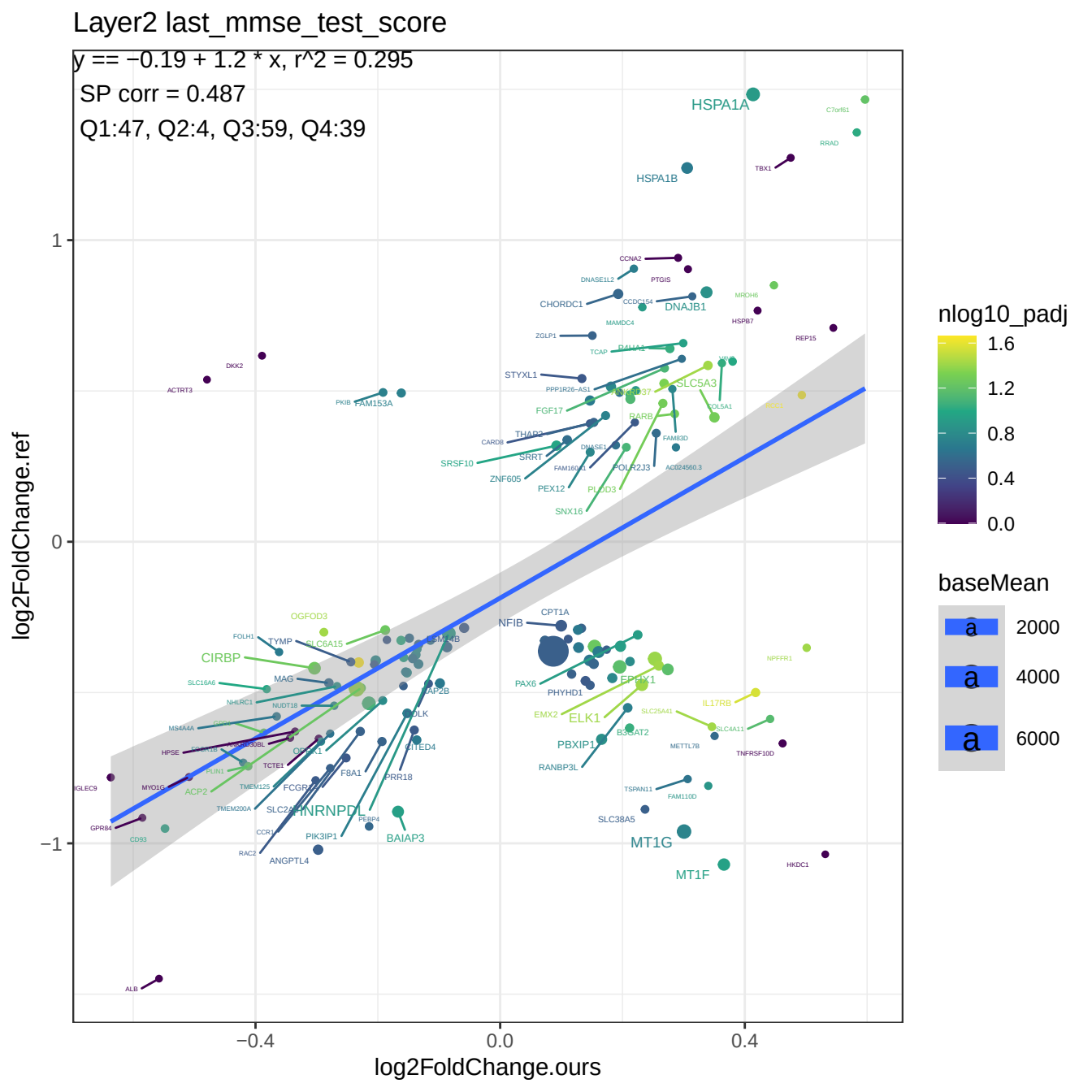
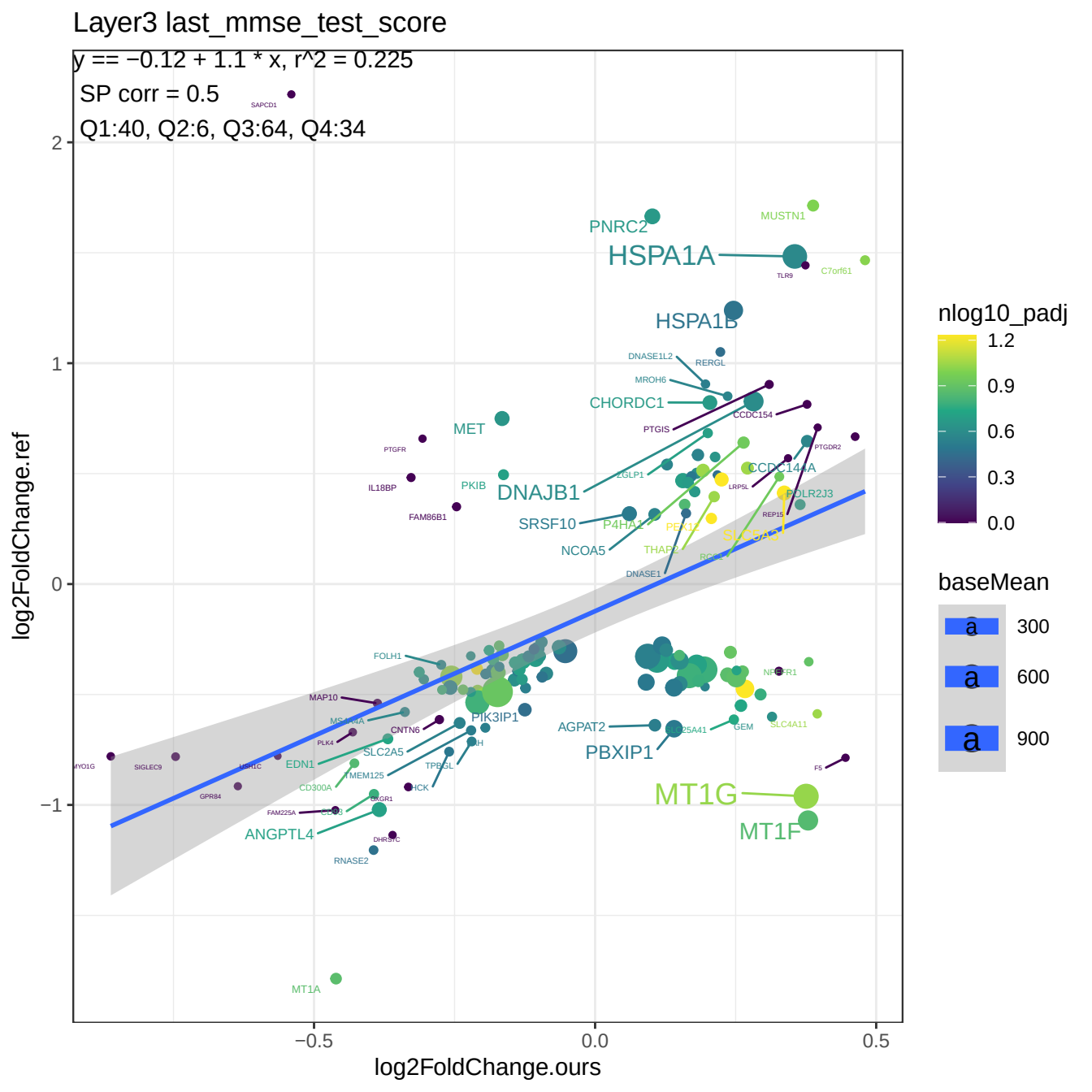
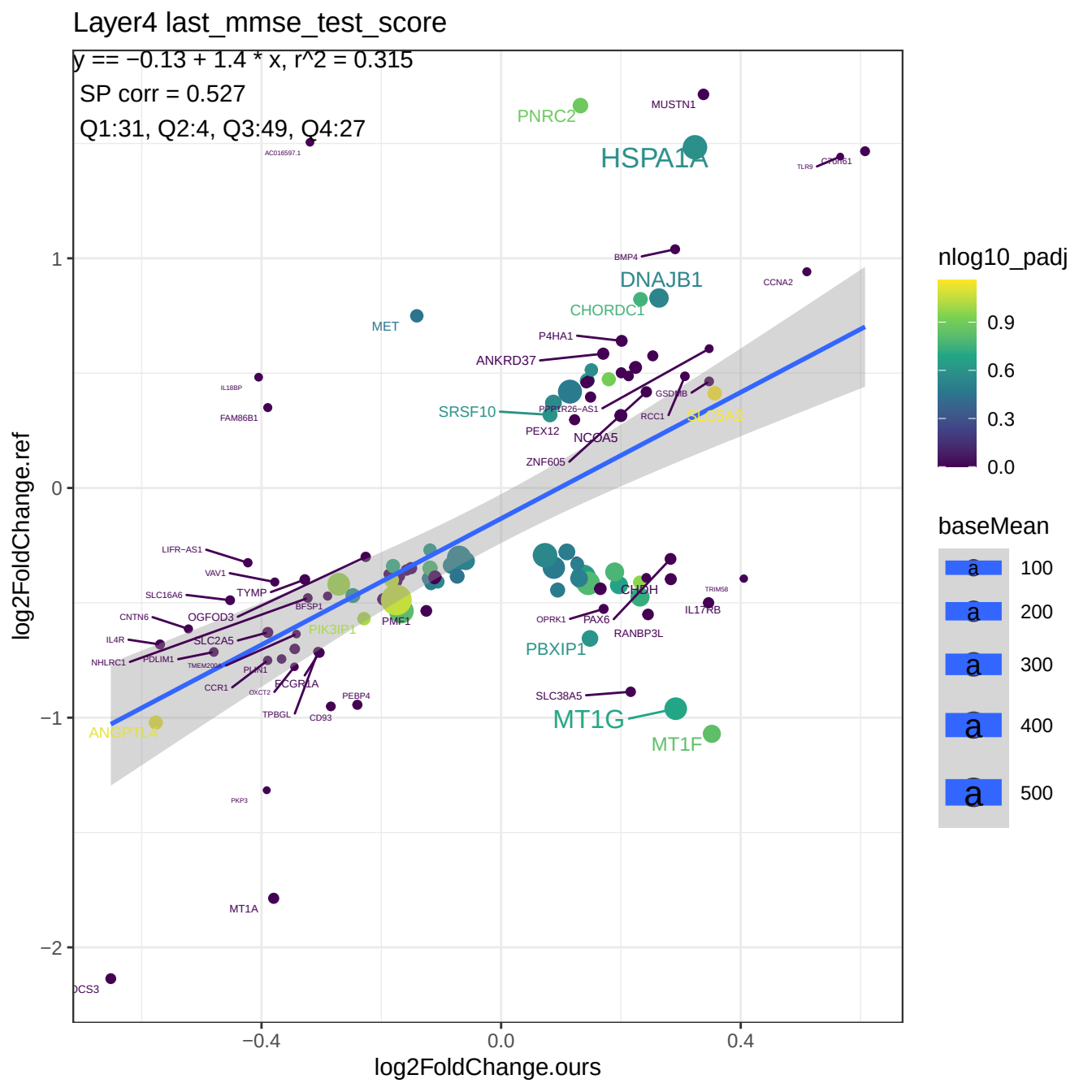
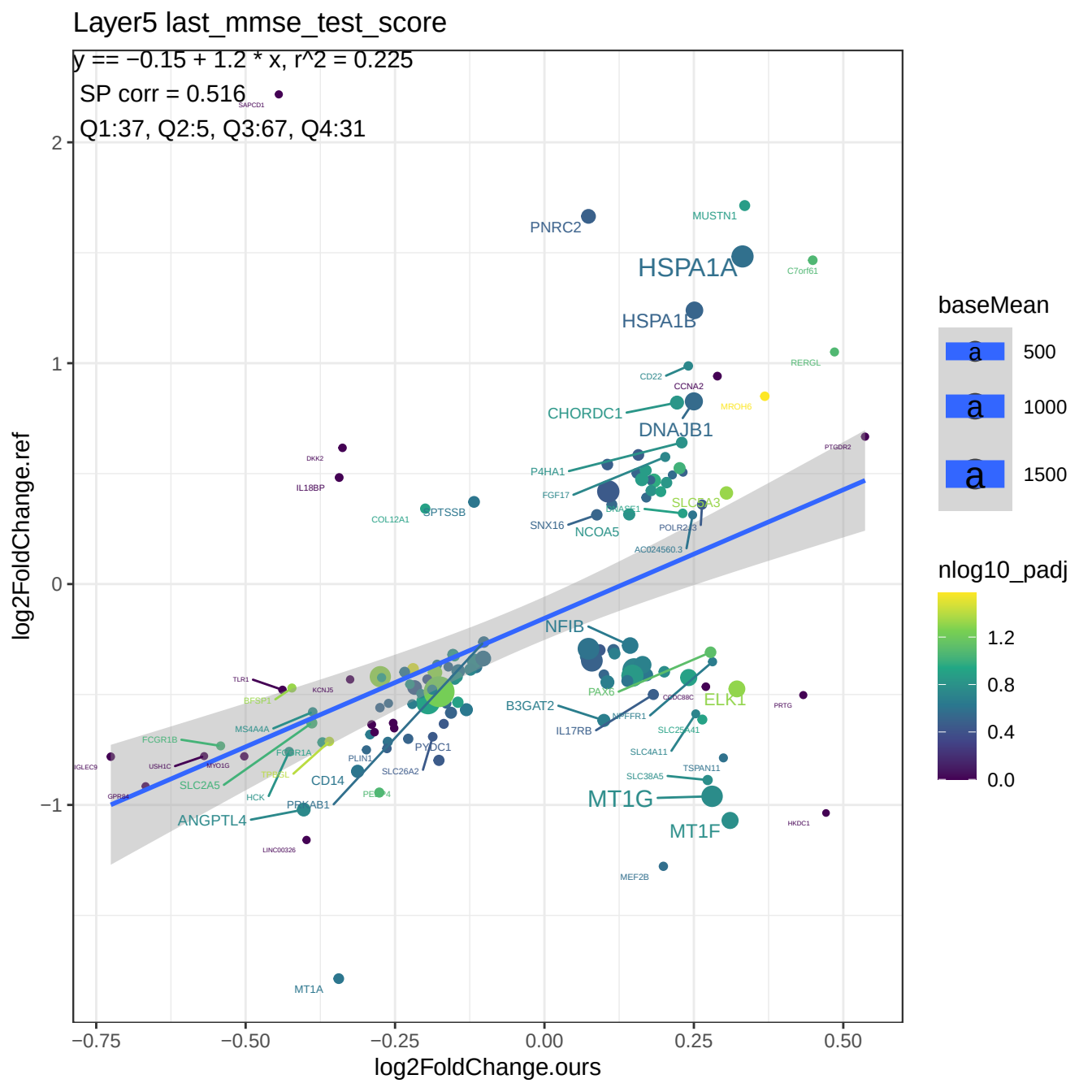
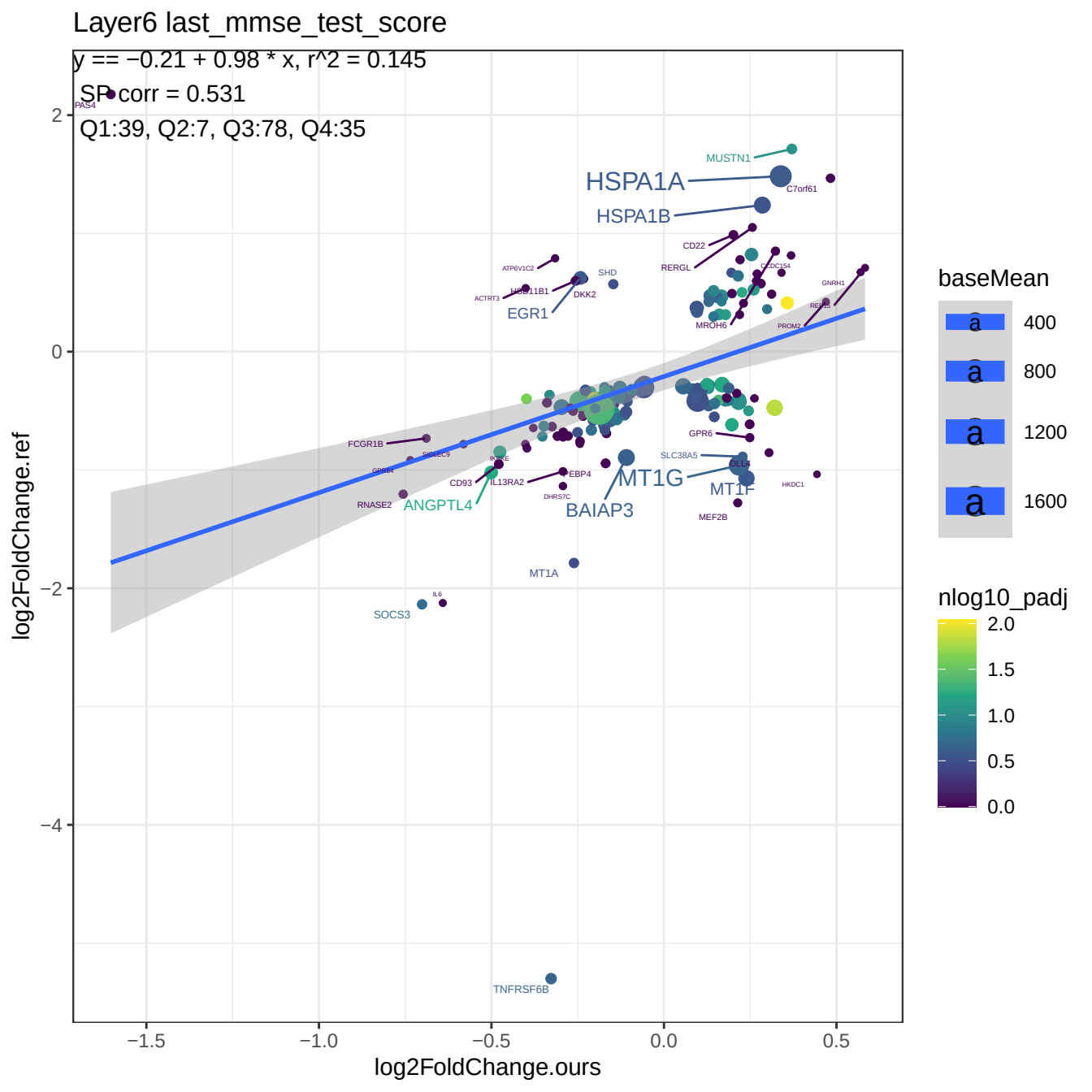
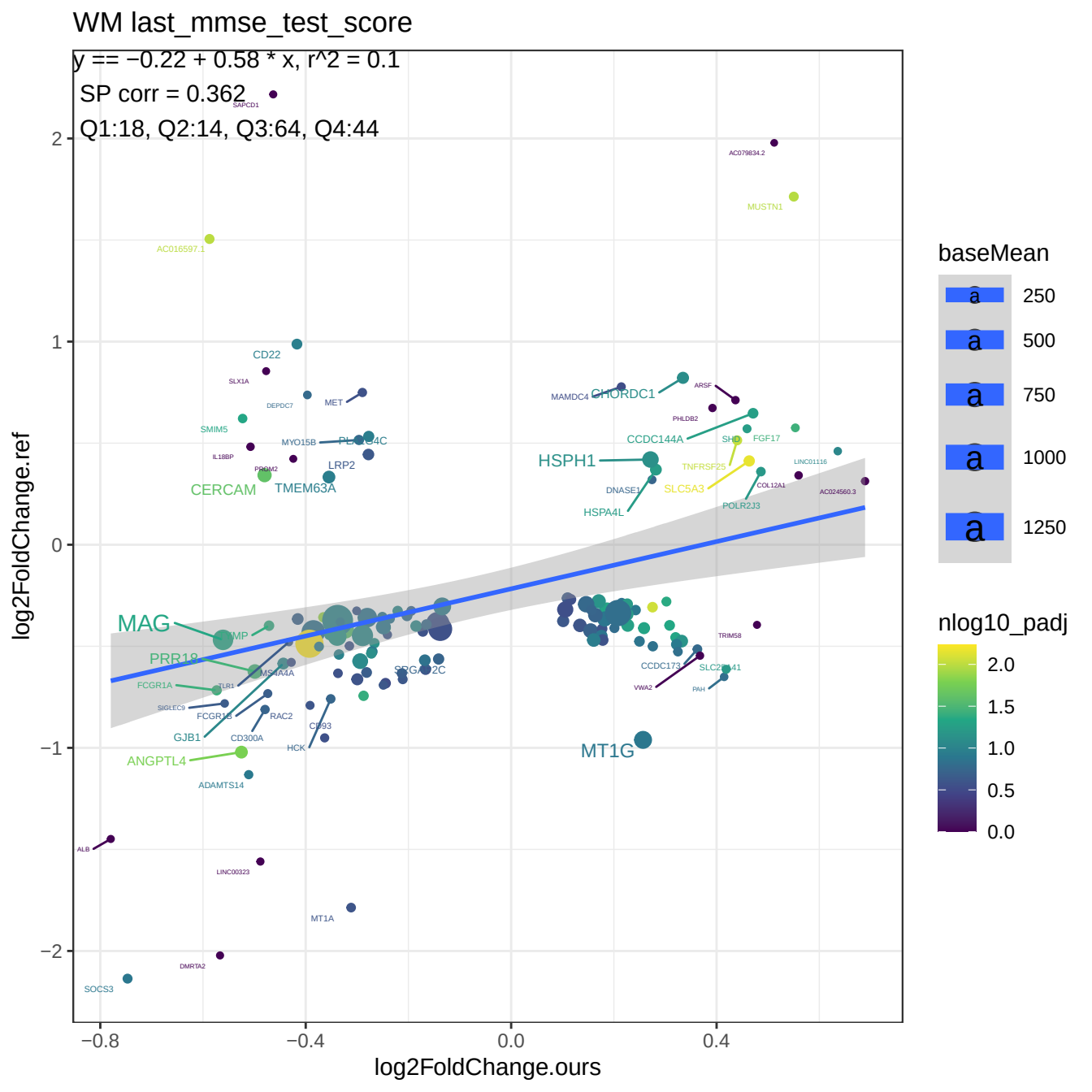

Layer2 last_mmse_test_score
y == −0.19 + 1.2 * x, r^2 = 0.295
SP corr = 0.487
Q1:47, Q2:4, Q3:59, Q4:39
log2FoldChange.ref
log2FoldChange.ours
1
0
−1
−0.4
0.0
0.4
HSPA1A
HSPA1B
C7orf61
RRAD
TBX1
CCNA2
DNASE1L2
PTGIS
CHORDC1
CCDC154
DNAJB1
MRO6
HSPB7
MAMDC4
ZGLP1
TCAP
REP15
DKK2
ACTRT3
PKIB
FAM153A
STYXL1
PPP1R26−AS1
SLC5A3
FGF17
RARB
FAM83D
CARD8
THAP3
DNASE1
SRRT
SRSF10
FAM160A1
POLR2J3
AC024560.3
ZNF605
PEX12
PLOD3
SNX16
OGFOD3
FOLH1
TYMP
SLC6A15
CIRBP
SLC16A6
MAG
NHERC1
NUDT18
MS4A4A
HPSE
TCTE1
PLIN1
MYO1G
ACP2
TMEM125
TMEM200A
FCGR2A
SLC2A6
GPR84
CD93
CCR1
PIK3IP1
RAC2
ANGPTL4
F8A1
PRR18
HNRNPDL
PEBP4
BAIAP3
CPT1A
NFIB
PAX6
PHYHD1
EMX2
ELK1
PBXIP1
RANBP3L
ERBB4
IL17RB
SLC25A41
SLC4A11
METTL7B
TNFRSF10D
NPFFR1
TSPAN11
FAM110D
SLC38A5
MT1G
MT1F
HKDC1
ALB
nlog10_padj
1.6
1.2
0.8
0.4
0.0
baseMean
a
2000
a
4000
a
6000


Layer3 last_mmse_test_score
y == −0.12 + 1.1 * x, r^2 = 0.225
SP corr = 0.5
Q1:40, Q2:6, Q3:64, Q4:34
log2FoldChange.ref
log2FoldChange.ours
2
1
0
−1
−0.5
0.0
0.5
nlog10_padj
1.2
0.9
0.6
0.3
0.0
baseMean
300
600
900
PNRC2
HSPA1A
HSPA1B
MUSTN1
C7orf61
TLR9
DNASE1L2
RERGL
MROH6
CHORDC1
CCDC154
PTGIS
PTGDR2
MET
PTGFR
IL18BP
PKIB
DNAJB1
ZGLP1
LRP5L
CCDC144A
POLR2J3
REP15
FAM86B1
SRSF10
P4HA1
PEX12
SLC25A13
NCOA5
THAP2
RGS19
DNASE1
FOLH1
NPFFR1
MAP10
PIK3IP1
AGPAT2
SLC25A41
GEM
SLC4A11
CNTN6
PLK4
SLC2A5
EDN1
TMEM125
TPBGL
CD300A
HCK
GPR84
FAM225A
CXCR1
ANGPTL4
DHRS3C
RNASE2
MT1A
MT1G
MT1F
PBXIP1
F5
SIGLEC9
MSH1C
MYO1G
SAPCD1


Layer4 last_mmse_test_score
y == −0.13 + 1.4 * x, r^2 = 0.315
SP corr = 0.527
Q1:31, Q2:4, Q3:49, Q4:27
log2FoldChange.ref
log2FoldChange.ours
nlog10_padj
0.9
0.6
0.3
0.0
baseMean
100
200
300
400
500
PNRC2
MUSTN1
HSPA1A
DNAJB1
CHORDC1
MET
BMP4
CCNA2
P4HA1
ANKRD37
GSDMB
RCC1
SRSF10
PEX12
NCOA5
ZNF605
SLC5A3
FAM86B1
IL18BP
LIFR−AS1
VAV1
TYMP
SLC16A6
BFSP1
CNTN6
OGFOD3
IL4R
SLC2A5
NHLRC1
PDLIM1
PLIN1
CCR1
FCGR1A
TPBGL
CD93
PEBP4
PMF1
PIK3IP1
ANGPTL4
PKP3
MT1A
CHDH
TRIM58
IL17RB
OPRK1
PAX6
RANBP3L
PBXIP1
SLC38A5
MT1G
MT1F


Layer5 last_mmse_test_score
y == −0.15 + 1.2 * x, r^2 = 0.225
SP corr = 0.516
Q1:37, Q2:5, Q3:67, Q4:31
log2FoldChange.ref
log2FoldChange.ours
2
1
0
−1
−0.75
−0.50
−0.25
0.00
0.25
0.50
baseMean
500
1000
1500
nlog10_padj
1.2
0.8
0.4
0.0
PNRC2
MUSTN1
HSPA1A
C7orf61
HSPA1B
RERGL
CD22
CCNA2
MROH6
CHORDC1
DNAJB1
PTGDR2
P4HA1
DKK2
IL18BP
FGF17
SLC6A3
COL12A1
GPTSSB
SNX16
NCOA5
POLR2J3
AC024560.3
NFIB
TLR1
KCNJ5
BFSP1
PAX6
ELK1
B3GAT2
PRTG
MS4A4A
IL17RB
FCGR1B
FCGR1A
PYDC1
SLC4A11
TSPAN11
SLC38A5
SLC2A5
CD14
PLIN1
SLC26A2
MT1G
HCK
PRKAB1
MT1F
ANGPTL4
HKDC1
LINC00326
MEF2B
MT1A


Layer6 last_mmse_test_score
y == −0.21 + 0.98 * x, r^2 = 0.145
SP corr = 0.531
Q1:39, Q2:7, Q3:78, Q4:35
log2FoldChange.ref
log2FoldChange.ours
2
0
−2
−4
−1.5
−1.0
−0.5
0.0
0.5
HSPA1A
HSPA1B
MUSTN1
C7orf61
CD22
RERGL
EGR1
DKK2
SHD
ATP6V1C2
ACTRT3
MROH6
PROM2
GNRH1
FCGR1B
SIGLEC9
GPR84
RNASE2
ANGPTL4
CD93
IL13RA2
EBP4
DHRS7C
BAIAP3
MT1G
MT1F
SLC38A5
GPR6
HKDC1
MEF2B
MT1A
SOCS3
IL6
TNFRSF6B
baseMean
a
400
800
1200
1600
nlog10_padj
2.0
1.5
1.0
0.5
0.0


WM last_mmse_test_score
y == −0.22 + 0.58 * x, r^2 = 0.1
SP corr = 0.362
Q1:18, Q2:14, Q3:64, Q4:44
log2FoldChange.ref
log2FoldChange.ours
baseMean
250
500
750
1000
1250
nlog10_padj
2.0
1.5
1.0
0.5
0.0
MAG
PRR18
ANGPTL4
GJB1
CERCAM
TMEM63A
HSPH1
CHORDC1
CCDC144A
HSPA4L
MT1G
CD22
MET
LRP2
MYO15B
SMIM5
SLX1A
DEPDC7
IL18BP
PRKM2
MAMDC4
ARSF
PHLDB2
SHD
FGF17
TNFRSF25
SLC5A3
DNASE1
COL12A1
POLR2J3
LINC01116
AC024560.3
TRIM58
CCDC173
SLC2A11
VWA2
PAH
MS4A4A
FCGR1A
TLR1
SIGLEC9
FCGR1B
RAC2
CD93
CD300A
HCK
ADAMTS14
SRGAP2C
AC079834.2
MUSTN1
AC016597.1
SAPCD1
ALB
LINC00323
MT1A
DMRTA2
SOCS3